Nov 3, 2026 (Due: 08:00 Nov 10, 2026)

1. Let $(\hat{\lambda}, \hat{x})$ be an approximate eigenpair of a Hermitian matrix A, and $r = A\hat{x} - \hat{x}\hat{\lambda}$ be the residual. Suppose that $\|\hat{x}\|_2 = 1$. Show that there exists an eigenvalue of A, denoted by λ, such that

$$|\lambda - \hat{\lambda}| \leq \|r\|_2.$$

2. Let $A \in \mathbb{C}^{m\times n}$, $X \in \mathbb{C}^{n\times n}$, $Y \in \mathbb{C}^{m\times m}$. Show that

$$|\sigma_i(Y^*AX) - \sigma_i(A)| \leq \epsilon \cdot \sigma_i(A),$$

where $\epsilon = \max\{\|X^*X - I_n\|_2, \|Y^*Y - I_m\|_2\}$

3. Given x, $y \in \mathbb{R}^n$. Describe in detail how to construct a rotation matrix Q such that the columns of $[x, y]Q$ are mutually orthogonal.

4. Create a full matrix with several nontrivial Jordan blocks using $A = XJX^{-1}$ where X is a randomly generated nonsingular matrix. Make observations on the forward error of the computed eigenvalues of A.

5. Have a quick glance at the paper "From Random Polygon to Ellipse: An Eigenanalysis" by A. N. Elmachtoub and C. F. Van Loan (available on eLearning). Reproduce the experiments in this paper and play with several variants mentioned in the lecture. You are encouraged to try other variants.

6. (optional) Use truncated SVD to compress some grayscale images. Visualize the results. If you only have colored images, you can convert them to grayscale using

$$\text{gray} = \alpha \cdot \text{red} + \beta \cdot \text{green} + \gamma \cdot \text{blue}.$$

Common choices of the constants are $(\alpha, \beta, \gamma) = (0.299, 0.587, 0.114)$ and $(\alpha, \beta, \gamma) = (0.2126, 0.7152, 0.0722)$.

You are also encouraged to think about how to compress colored images.

7. (optional) Let A, $E \in \mathbb{C}^{n\times n}$ be Hermitian matrices. Suppose that $x(\theta) \in \mathbb{C}^n$ is the normalized eigenvector of $A + \theta E$ corresponding to the eigenvalue $\lambda(\theta)$, and both $\lambda(\theta)$ and $x(\theta)$ are smooth functions in θ. Show that

$$\frac{\mathrm{d}\lambda(\theta)}{\mathrm{d}\theta} = x(\theta)^* E x(\theta)$$

if $\lambda(\theta)$ is a simple eigenvalue.

8. (optional) Show that small componentwise relative perturbations to a bidiagonal matrix lead to small relative perturbations to its singular values.

Hint: Represent the componentwise relative perturbations using diagonal scaling.